
Web Technologies
IT361

Project

Deadline: Monday, 4/5/2026 @ 23:59

[Total Mark for this Assignment is 14]

Students Details:

CRN: ###

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Learning Outcome(s):

2. Create web pages using HTML5 and CSS3.

3. Develop dynamic web pages using JavaScript.

6. Build web applications using PHP and MySQL.

Project

(14 Marks)

In this project, you need to build a website **based on one of the suggested project ideas below**. Your project should be from one of the ideas below. You can work on it individually or as a group of a maximum of 3 members.

Suggested Project ideas

1. **Online Course Registration System:** Home page, Users register/login page, course listing page, course enrollment page, and enrollment history page.
2. **Hotel Booking Website:** Home page, Users register/login page, room search page, room details page, and booking history page.
3. **Online Shopping Website:** Home page, Users register/login page, product categories page, shopping cart page, and order history page.
4. **Clinic Appointment System:** Home page, Users register/login page, doctors listing page, appointment booking page, and appointment history page.

The website must cover all the following aspects of the course (HTML, CSS, JavaScript, PHP, MySQL).

Technology	Description	Mark
HTML	HTML Web elements (header, footer, list, table, form ...)	4
	Navigation (Links are clear and located in the same area on each page. All links are functional)	
CSS	Design (using CSS) Website demonstrates design principles: with consistent alignment, proximity, repetition, and contrast...	3
JavaScript	Form validation, input/output	2
	Create and populate a database in MySQL (1-2 tables).	5

PHP and MySQL	Select records from MySQL database using PHP	
	Insert records into MySQL database using PHP	
	Update records into MySQL database using PHP	
	Delete records from MySQL database using PHP	
Total		14

Important notes:

- The number of Web pages within the website must be **at least 3 pages**.
- The number of dynamic Web pages (using JavaScript) must be **at least 2 pages**.
- The Website must connect to a database (**1 Select + 1 Insert + 1 Update and 1 Delete queries**) and take a screenshot for each one.
- For form validation, at least one popup message must appear after the submit button has been clicked. The message must contain a list of all validation errors on the Web page.
- The project needs to be **submitted in BB** (as an assignment) by **ONE** of the group members. Your instructor will create the submission link.
- Your HTML, CSS, JavaScript, PHP and MySQL codes/scripts that you paste into your answer part of this file must be editable, **not screenshots**.
- You need to take screenshots from each output pages and paste them into your answer in this file.
- Your project must be **original work**. The project will be checked using a **plagiarism tool**. Any matching with an **online project, AI work** or **another/previous student project** will be given **ZERO**.
- **The maximum** number of students in the group is **3**. Only the **group leader** will **upload** the final report on the **blackboard**.

Presentation:

- Each group needs to make a presentation OR a video recording (based on your instructor) of their project. The presentation date will be announced later by your instructor.

- The maximum time for a presentation is **5 mins**.
- In a presentation case, **All** the group members need to **present**.
- In both presentation and video recording cases, **no** need to make **PowerPoint slides**. You need to **run/demonstrate** your project pages and present their functionality and interfaces/pages.

Files to submit/upload:

1. **IT361_Project.docx:**

A word file of the project that contains your answer.

2. **IT361_Project.pdf**

A PDF file of the project that contains your answer.

3. **IT361_Project.zip:**

Contains all the coding files of the project (HTML, CSS, images JavaScript, PHP and MySQL).

*Learning
Outcome(s):*

*2. Create web
pages using
HTML5 and
CSS3.*

HTML

(4 Marks)

Project Title: Clinic Appointment System

Section 1: index.html (Home Page Implementation)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Home - Clinic Appointment System</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Welcome to Our Specialized Clinic</h1>
```

```
<nav>
```

```
<ul>
```

```
<li><a href="index.html">Home</a></li>
```

```
<li><a href="doctors.html">Doctors</a></li>
```

```
<li><a href="booking.html">Book Appointment</a></li>
```

```
</ul>
```

```
</nav>
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<h2>About Our Clinic</h2>
```

```
<p>We provide the best healthcare services for our patients using the latest  
medical technologies.</p>
```

```
</section>
```

```
<section>
```

```
<h2>Our Services</h2>
```

```
<ul> <li>General Medical Consultations</li>
```

```
<li>Comprehensive Periodic Checkups</li>

<li>24/7 Emergency Care</li>

</ul>

</section>

</main>

<footer>

<p>All Rights Reserved &copy; 2026 - IT361 Project</p>

</footer>

</body>

</html>
```

Output: Home page showing the navigation menu and a welcoming header for the clinic.

Welcome to Our Specialized Clinic

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

About Our Clinic

We provide the best healthcare services for our patients using the latest medical technologies.

Our Services

- General Medical Consultations
- Comprehensive Periodic Checkups
- 24/7 Emergency Care

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Section 2: doctors.html (Doctors List Implementation)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Doctors - Clinic Appointment System</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Our Professional Doctors</h1>
```

```
<nav>
```

```
<ul>
```

```
<li><a href="index.html">Home</a></li>
```

```
<li><a href="doctors.html">Doctors</a></li>
```

```
<li><a href="booking.html">Book Appointment</a></li>
```


</nav>

</header>

<main>

<h2>Available Specialists</h2>

<table border="1">

<thead>

<tr>

<th>Doctor Name</th>

<th>Specialization</th>

<th>Working Hours</th>

</tr>

</thead>

<tbody>

<tr>

<td>Dr. Ahmed Ali</td>

<td>Pediatrics</td>

<td>8:00 AM - 2:00 PM</td>

</tr>

<tr>

<td>Dr. Sara Mohammed</td>

<td>General Medicine</td>

<td>2:00 PM - 8:00 PM</td>

</tr>

<tr>

<td>Dr. Khalid Mansour</td>

<td>Cardiology</td>

<td>9:00 AM - 3:00 PM</td>

</tr>

<tr>

<td>Dr. Nora Al-Otaibi</td>

<td>Dermatology</td>

```
<td>4:00 PM - 10:00 PM</td>

</tr>

</tbody>

</table>

</main>

<footer>

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</footer>

</body>

</html>
```

Output: Doctors page featuring an HTML table that lists the names, specializations, and working hours of our 4 specialists.

Our Professional Doctors

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

Available Specialists

Doctor Name	Specialization	Working Hours
Dr. Ahmed Ali	Pediatrics	8:00 AM - 2:00 PM
Dr. Sara Mohammed	General Medicine	2:00 PM - 8:00 PM
Dr. Khalid Mansour	Cardiology	9:00 AM - 3:00 PM
Dr. Nora Al-Otaibi	Dermatology	4:00 PM - 10:00 PM

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Section 3: booking.html (Booking Form Implementation)

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<title>Book Appointment - Clinic Appointment System</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Book Your Appointment Now</h1>
```

```
<nav>
```

```
<ul>
```

```
<li><a href="index.html">Home</a></li>
```

```
<li><a href="doctors.html">Doctors</a></li>
```

```
<li><a href="booking.html">Book Appointment</a></li>
```

```
</ul>
```

```
</nav>
```

```
</header>
```

```
<main>
```

```
<h2>Appointment Form</h2>
```

```
<form id="appointmentForm" action="process_booking.php" method="POST">
```

```
<label for="patientName">Patient Name:</label><br>
```

```
<input type="text" id="patientName" name="patientName"><br><br>
```

```
<label for="doctor">Choose Doctor:</label><br>
```

```
<select id="doctor" name="doctor">
```

```
<option value="">-- Select a Doctor --</option>
```

```
<option value="ahmed">Dr. Ahmed Ali (Pediatrics)</option>
```

```
<option value="sara">Dr. Sara Mohammed (General Medicine)</option>
```

```
<option value="khalid">Dr. Khalid Mansour (Cardiology)</option>
```

```
<option value="nora">Dr. Nora Al-Otaibi (Dermatology)</option>
```

```
</select><br><br>
```

```
<label for="appDate">Appointment Date:</label><br>
```

```
<input type="date" id="appDate" name="appDate"><br><br>
```

```
<input type="submit" value="Submit Request">
```

```
</form>
```

```
</main>
```

```
<footer>
```

```
<p>All Rights Reserved &copy; 2026 - IT361 Project</p>
```

```
</footer>
```

```
</body>
```

```
</html>
```

Output: Appointment booking form with various input types (text, date, and dropdown list).

And Dropdown list showing the four available doctors for selection

Book Your Appointment Now

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

Appointment Form

Patient Name:

Choose Doctor:

Appointment Date:

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Book Your Appointment Now

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

Appointment Form

Patient Name:

Choose Doctor:

Dr. Ahmed Ali (Pediatrics)

Dr. Sara Mohammed (General Medicine)

Dr. Khalid Mansour (Cardiology)

Dr. Nora Al-Otaibi (Dermatology)

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Book Your Appointment Now

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

Appointment Form

Patient Name:

Choose Doctor:

Appointment Date:

April 2026

Su	Mo	Tu	We	Th	Fr	Sa
29	30	31	1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	1	2
3	4	5	6	7	8	9

Clear Today

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Book Your Appointment Now

- [Home](#)
- [Doctors](#)
- [Book Appointment](#)

Appointment Form

Patient Name:

Choose Doctor:

Appointment Date:

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*Learning
Outcome(s):*

*2. Create web
pages using
HTML5 and
CSS3.*

CSS

(**3 Marks**)

2.Styles.css:

```
/* style.css - Clinic Appointment System */
```

```
/* Reset basic styles */
```

```
* {  
  
    margin: 0;  
  
    padding: 0;  
  
    box-sizing: border-box;  
  
    font-family: Arial, sans-serif;  
  
}
```

```
/* Body styles */
```

```
body {  
  
    background-color: #f5f5f5;  
  
    color: #333;
```

```
line-height: 1.6;

}

/* Header styles */

header {

    background-color: #4CAF50;

    color: white;

    padding: 20px 0;

    text-align: center;

}

header h1 {

    font-size: 2em;

    margin-bottom: 10px;

}
```

```
/* Navigation styles */
```

```
nav ul {
```

```
    list-style: none;
```

```
    display: flex;
```

```
    justify-content: center;
```

```
    gap: 20px;
```

```
}
```

```
nav ul li a {
```

```
    color: white;
```

```
    text-decoration: none;
```

```
    font-weight: bold;
```

```
    padding: 8px 12px;
```

```
    transition: background-color 0.3s, border-radius 0.3s;
```

```
}
```

```
nav ul li a:hover {  
  
    background-color: #45a049;  
  
    border-radius: 5px;  
  
}
```

```
/* Main content */  
  
main {  
  
    padding: 20px;  
  
    max-width: 1000px;  
  
    margin: 0 auto;  
  
}
```

```
/* Sections */  
  
section {  
  
    margin-bottom: 30px;  
  
    padding: 15px;
```

```
background-color: white;

border-radius: 8px;

box-shadow: 0px 2px 5px rgba(0,0,0,0.1);

}
```

```
section h2 {

margin-bottom: 15px;

color: #4CAF50;

}
```

```
/* Lists inside sections */
```

```
ul {

padding-left: 20px;

}
```

```
ul li {
```

```
margin-bottom: 8px;

}

/* Table styles for doctors.html */

table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;

}

table th, table td {

    border: 1px solid #ddd;

    padding: 12px;

    text-align: left;

}
```

```
table th {
```

```
    background-color: #4CAF50;
```

```
    color: white;
```

```
}
```

```
table tr:nth-child(even) {
```

```
    background-color: #f2f2f2;
```

```
}
```

```
/* Form styles for booking.html */
```

```
form {
```

```
    background-color: white;
```

```
    padding: 20px;
```

```
    border-radius: 8px;
```

```
    box-shadow: 0px 2px 5px rgba(0,0,0,0.1);
```

```
}
```

```
form label {
```

```
    font-weight: bold;
```

```
}
```

```
form input[type="text"],
```

```
form input[type="date"],
```

```
form select {
```

```
    width: 100%;
```

```
    padding: 8px;
```

```
    margin-top: 5px;
```

```
    margin-bottom: 15px;
```

```
    border-radius: 5px;
```

```
    border: 1px solid #ccc;
```

```
}
```

```
form input[type="submit"] {  
  
    background-color: #4CAF50;  
  
    color: white;  
  
    padding: 10px 20px;  
  
    border: none;  
  
    border-radius: 5px;  
  
    cursor: pointer;  
  
    font-size: 1em;  
  
    transition: background-color 0.3s;  
  
}
```

```
form input[type="submit"]:hover {  
  
    background-color: #45a049;  
  
}
```

```
/* Footer styles */
```

```
footer {  
  
    background-color: #333;  
  
    color: white;  
  
    text-align: center;  
  
    padding: 15px 0;  
  
    margin-top: 40px;  
  
    font-size: 0.9em;  
  
}
```

*Learning
Outcome(s):*

*3. Develop
dynamic web
pages using
JavaScript.*

JavaScript

(2 Marks)

Section 3: JavaScript Form Validation

1. Description:

In this part, JavaScript is used to make the appointment form more interactive by checking the inputs before submission. It makes sure that the user cannot submit the form if any required field, such as the patient name, doctor selection, or appointment date, is left empty. If there are missing fields, a popup message will appear showing all the errors so the user can fix them.

2. JavaScript Code (script.js):

```
// This function checks the appointment form before it gets submitted
```

```
document.getElementById('appointmentForm').addEventListener('submit',  
function(event) {
```

```
    // We use this array to collect any error messages
```

```
    let errors = [];
```

```
    // Get the values entered by the user
```

```
    const patientName = document.getElementById('patientName').value.trim();
```

```
    const doctor = document.getElementById('doctor').value;
```

```
const appDate = document.getElementById('appDate').value;

// Check if the patient name is empty
if (patientName === "") {
    errors.push("- Please enter the Patient Name.");
}

// Check if the user selected a doctor
if (doctor === "") {
    errors.push("- Please select a Doctor from the list.");
}

// Check if the appointment date is provided
if (appDate === "") {
    errors.push("- Please select a valid Appointment Date.");
}

// If there are any errors, stop the form and show them in a popup
if (errors.length > 0) {
    event.preventDefault(); // Prevent sending the form to PHP
```

```
alert("Validation Errors:\n" + errors.join("\n"));

}

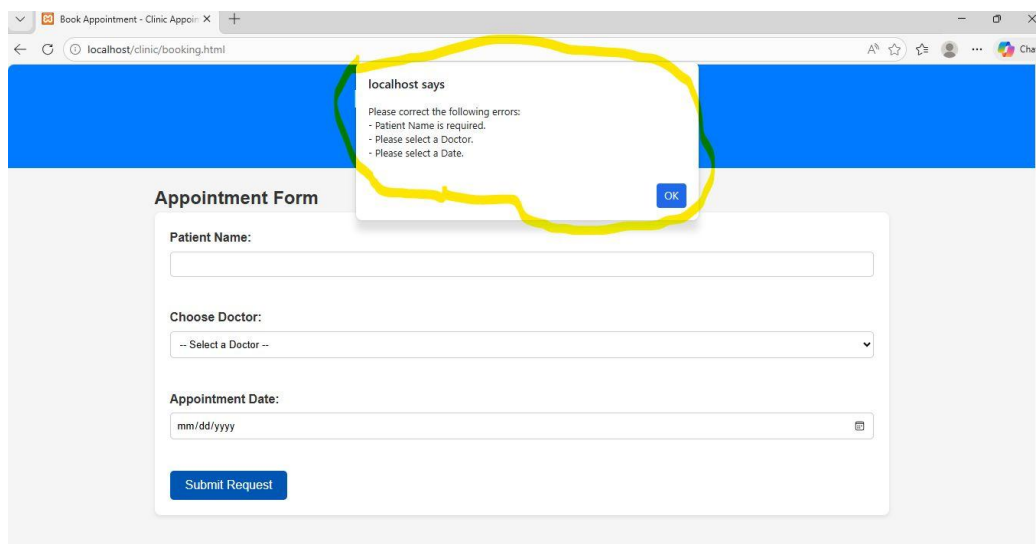
});
```

3. Updated HTML Linkage (booking.html):

The following line was added at the end of the `booking.html` file, right before the closing `</body>` tag, to connect the JavaScript file to the page:

```
<script src="script.js"></script>
```

4. Output (Screenshot):



*Learning
Outcome(s):*

*6. Build web
applications using
PHP and MySQL.*

PHP + MySQL

(5 Marks)

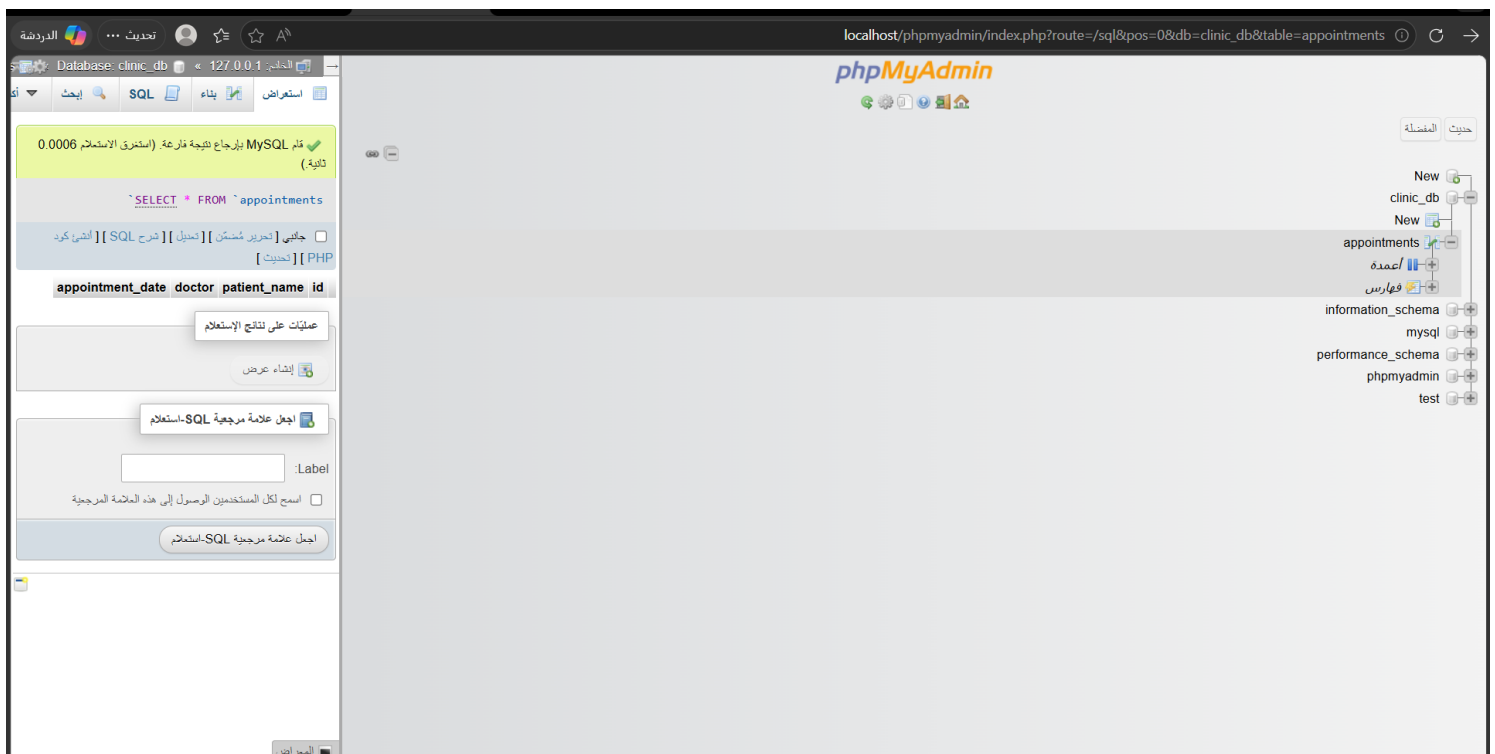
1.Database Creation

A database named clinic_db was created, and a table called appointments was created to store patient appointment information.

```
CREATE DATABASE clinic_db;
```

```
USE clinic_db;
```

```
CREATE TABLE appointments (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    patient_name VARCHAR(100),  
    doctor VARCHAR(100),  
    appointment_date DATE  
);
```



2. Insert Record

A new appointment was successfully inserted into the database for the patient **Amani** with the doctor **Dr. Nora** on **2026-04-24**.

SQL:

```
INSERT INTO appointments (id, patient_name, doctor, appointment_date)
VALUES ('1', 'Amani', 'Dr. Nora', '2026-04-24');
```

PHP:

```
<?php
$conn = new mysqli("localhost", "root", "", "clinic_db");

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

$patientName = $_POST['patientName'];
$doctor = $_POST['doctor'];
$appDate = $_POST['appDate'];
```

```
$sql = "INSERT INTO appointments (patient_name, doctor, appointment_date)
VALUES ('$patientName', '$doctor', '$appDate')";
```

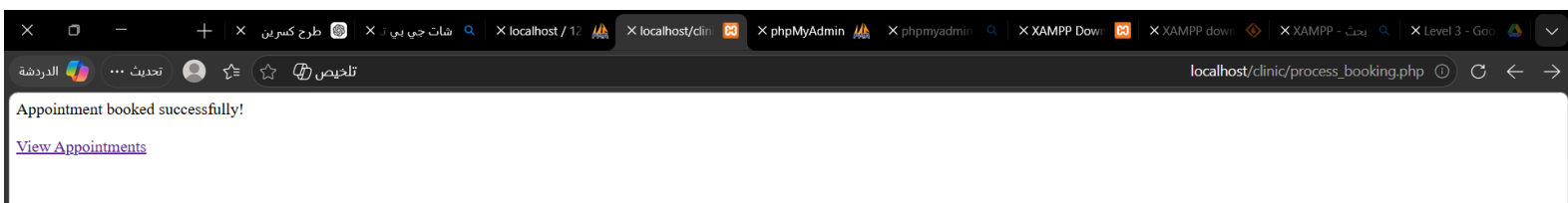
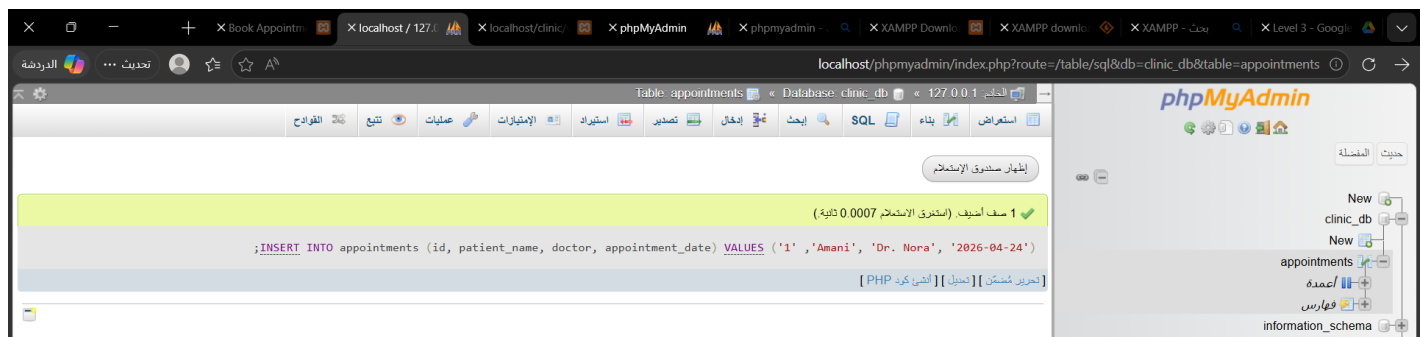
```
$conn->query($sql);
```

```
echo "Appointment booked successfully!<br>";
```

```
echo "<a href='view_appointments.php'>View Appointments</a>";
```

```
$conn->close();
```

```
?>
```



3. Select Records

The stored appointments were retrieved and displayed from the database using PHP, showing the patient name, doctor, and appointment date.

```
$conn = new mysqli("localhost", "root", "", "clinic_db");
```

```
$sql = "SELECT * FROM appointments";
```

```
$result = $conn->query($sql);
```

```
echo "<h2>Appointments List</h2>";
```

```
echo "<table border='1'>
```

```
<tr>
```

```
<th>ID</th>
```

```
<th>Patient Name</th>
```

```
<th>Doctor</th>
```

```
<th>Date</th>
```

```
<th>Actions</th>
```

```
</tr>";
```

```
while($row = $result->fetch_assoc()) {
```

```
echo "<tr>
```

```
<td>".$row["id"]."</td>
```

```
<td>".$row["patient_name"]."</td>
```

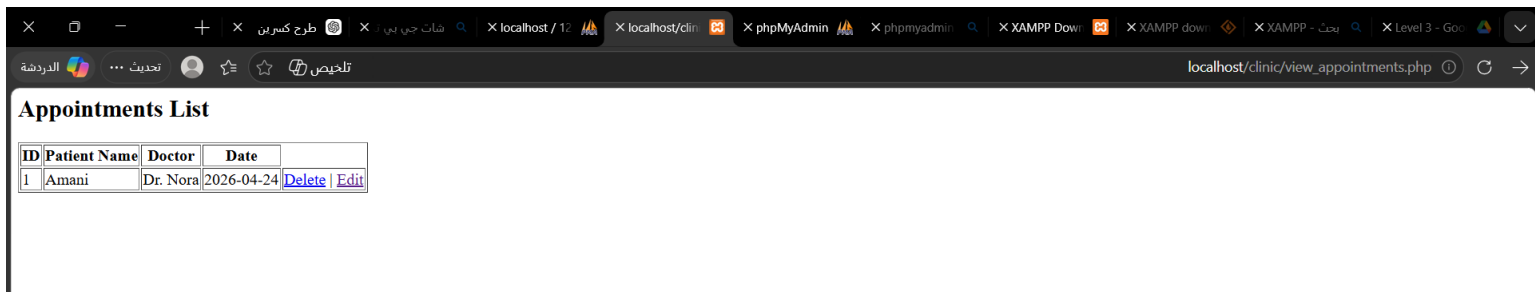
```
<td>".$row["doctor"]."</td>
```

```

<td>".$row["appointment_date"]."</td>
<td>
<a href='delete.php?id=".$row["id"]." '>Delete</a> |
<a href='edit.php?id=".$row["id"]." '>Edit</a>
</td>
</tr>";
}

echo "</table>";

```



4.Update Record

The appointment date was updated from **2026-04-24** to a new date.

SQL Code:

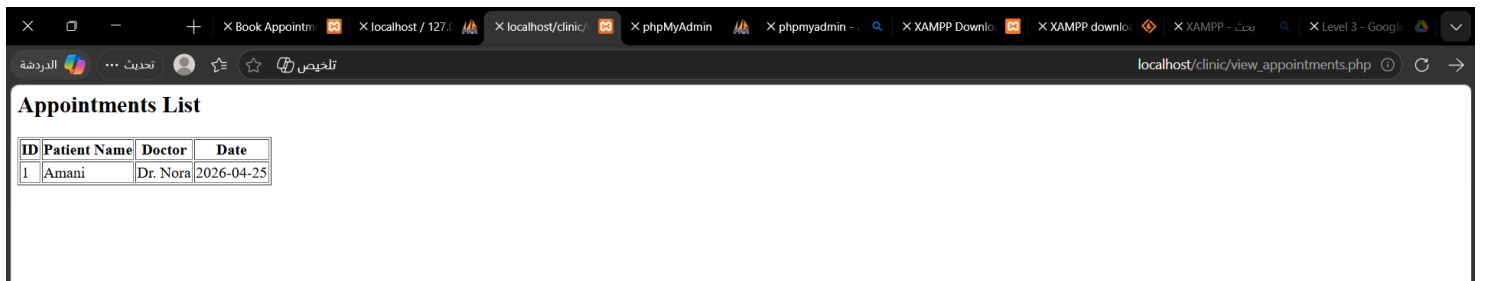
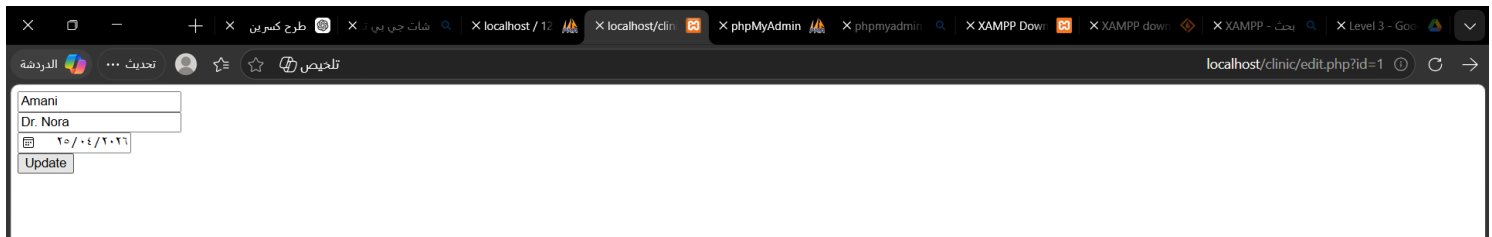
```
UPDATE appointments
SET appointment_date = '2026-04-25'
WHERE id = 1;
```

PHP Code:

```
while($row = $result->fetch_assoc()) {
    echo "<tr>
    <td>".$row["id"]."</td>
    <td>".$row["patient_name"]."</td>
    <td>".$row["doctor"]."</td>
    <td>".$row["appointment_date"]."</td>
    <td>
        <a href='delete.php?id=".$row["id"]."'">Delete</a> |
        <a href='edit.php?id=".$row["id"]."'">Edit</a>
    </td>
    </tr>";
}
```



The date has been modified by the patient.



5. Delete Record

The appointment record was successfully deleted from the database.

SQL Code:

```
DELETE FROM appointments  
WHERE id = 1;
```

PHP Code:

```
<?php  
  
$conn = new mysqli("localhost", "root", "", "clinic_db");  
  
$id = $_GET['id'];  
  
$sql = "DELETE FROM appointments WHERE id=$id";  
  
$conn->query($sql);  
  
header("Location: view_appointments.php");  
  
?>
```

